Antenna Theory

112-BOLCEW09

Learning Step Activities

- Understand Antenna Fundamentals
- Understand Induction and radiation fields
- Define Antenna Polarization
- Understand radiation patterns and directionality
- Contrast different Antenna Types

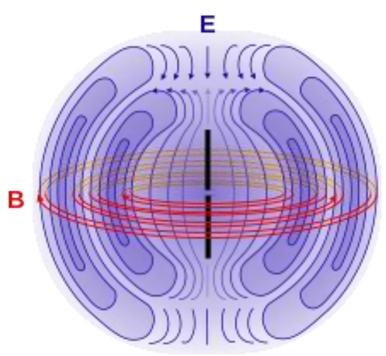
What Is An Antenna:

The interface between electromagnetic waves propagating through space and electric currents moving through system components

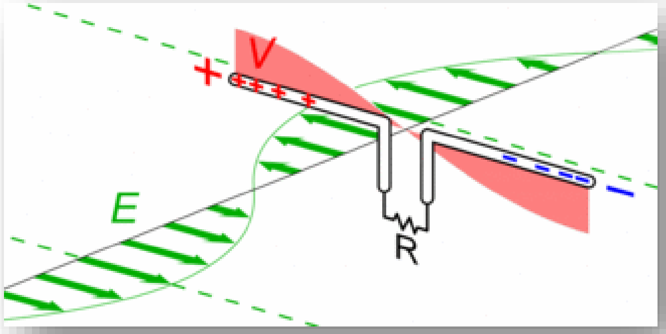
How it works:

Transmitter: Supplies an amplified electric current to the antenna's terminals causing the antenna to radiate energy in the form of EM waves

Receiver: Intercepts some of the EM waves, producing an electric current at the receiving antenna. The received electrical current is then carried to and amplified by the receiving device.



Electric Field
Magnetic Field
Current
Element



Fundamentals

The Principle of Induction

Recall Key Concepts:

- Electric and Magnetic Fields are linked
- A changing Electric Field creates a Changing Magnetic Field
- Changing Electric and Magnetic Fields create forces on charged particles (push or pull electrons)
- A current is collection of moving charged particles (electrons)

New Information

- A changing magnetic field creates a force on a wire
- The Force causes electrons to move, creating current in the wire
- Inversely, a current in a wire will also produce a magnetic field around the wire
- **This is called Induction and is the key concept in antennas!**

Induction in Action

A: **No Current** in the wire so **no magnetic field**

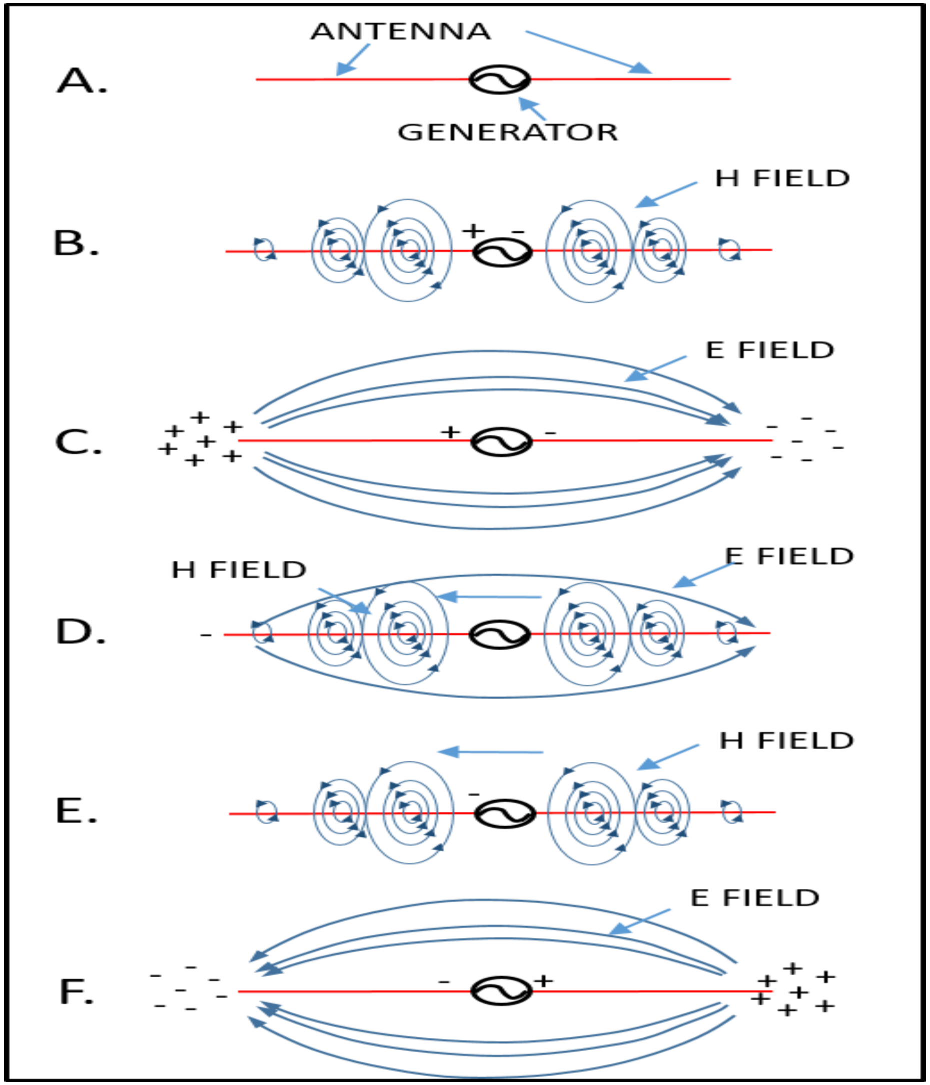
B: The generator is switched on and **current** begins running which creates a **magnetic field**

C. Both ends of the antenna are charged and the **current is no longer moving**, so there is **no magnetic field**

D: **Current** begins to reverse as the generator cycles which creates a new inverse **magnetic field**

E: **No Electric field** because the charges are at the middle, so **all energy** is in the **Magnetic field**

F: Both ends of the antenna are shared and the **current is no longer moving**, so there is **no magnetic field**



Antennas and EM Fields

- For an antenna to work properly, the following must exist:
  - Alternating electric energy emitted or received from a source
  - A good conductive material that easily allows current to flow
  - Generation of both electric and magnetic fields in the space surrounding the antenna

- **An Antenna will emit two types of fields**

  -Induction Field (near field)

  -Radiation Field (far field)

- If designed correctly, most of the energy received by the antenna will travel in the form of electromagnetic (EM) radiation

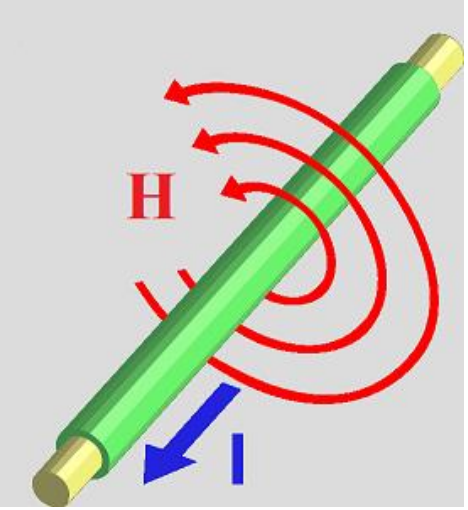
Induction Field

Induction Field (near field):

- The production of a changing magnetic field just outside an electrical conductor with an alternating current running in it.

No energy from this field is radiated or transmitted by the antenna

- This field is considered a localized field
- DOES NOT transmit EM energy
- Responsible for resonation effects that the antenna reflects to the generator

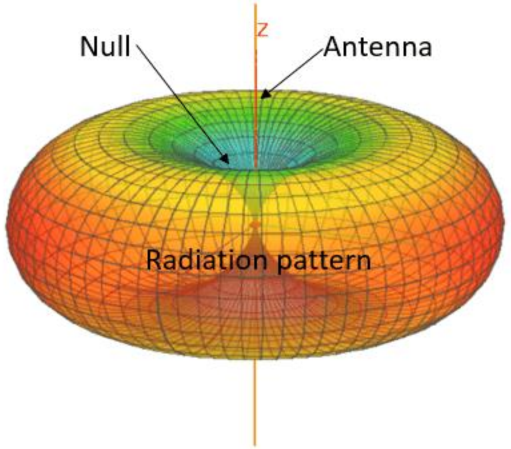


Radiation Field

Radiation Field definition(far field):

- **Responsible for the radiation and transmission of energy from the antenna!**
- **Decides the coverage pattern and directional strength of the antenna**
- Is composed the EM Waves emitted during transmissions
- Generally considered to be the area just past a few feet from the antenna and beyond

Key item: Radiation field decreases as the distance from the antenna is increased



Electrical Impedance?

Impedance is:

- The total opposition of an alternating current in a conductor.
- The combined forces of resistance and effects of magnetic induction.
- Expressed as the ration of the voltage and the current.

$$\boldsymbol{Z = \frac{E}{I}}$$

Impedance and antennas:

- Extremely important for antenna design!
- Antennas require alternating current to function.
- Incorrect impedance can cause low performance or even damage to radio equipment.

Resonance?

Resonance: An antennas enhanced ability to efficiently emit and receive energy at certain frequencies.

When does resonance occurs when:

- The phase of the alternating current in the antenna is shifted ½ of the phase of the induction field outside the antenna.
- When the Antenna is **½ the wavelength of the target frequency**
- Resonance can occur to a lesser extent at any number multiplied half the wavelength i.e. (1, ¼, 1/8, 1/16 wavelengths)
- Resonance is extremely important as its chief factor in deciding the design and purpose of an antenna

Resonance and Antenna Length

- **Ressonace is achieved at ½ the wavelength of a target frequency**
- The frequency and wavelength are inversely proportional
- The smaller the frequency the larger the antenna (think HF)
- The larger the frequency the smaller the antenna (think cell phone)

- Find the resonate frequency by dividing the speed of light by twice the target frequency

$$\mathrm{L} = \frac{C}{2f}$$

Where C=3x10^8 m/s

Where f= Target frequency in MHz

Standing Wave Ratio

Standing Wave Ratio:

- The ratio of outgoing energy vs. energy that is reflected back at a transmitter
- Can cause damage and drastically affect radio performance if too high.
- Is directly related to impedance, antenna resonance, and power output of the radio.

BLUF:

- Antennas need to be matched to the radio and the frequency!
- Don’t exceed power output of antennas!
- Be careful what kind of wire you use and how much of it you use!

Applied Knowledge

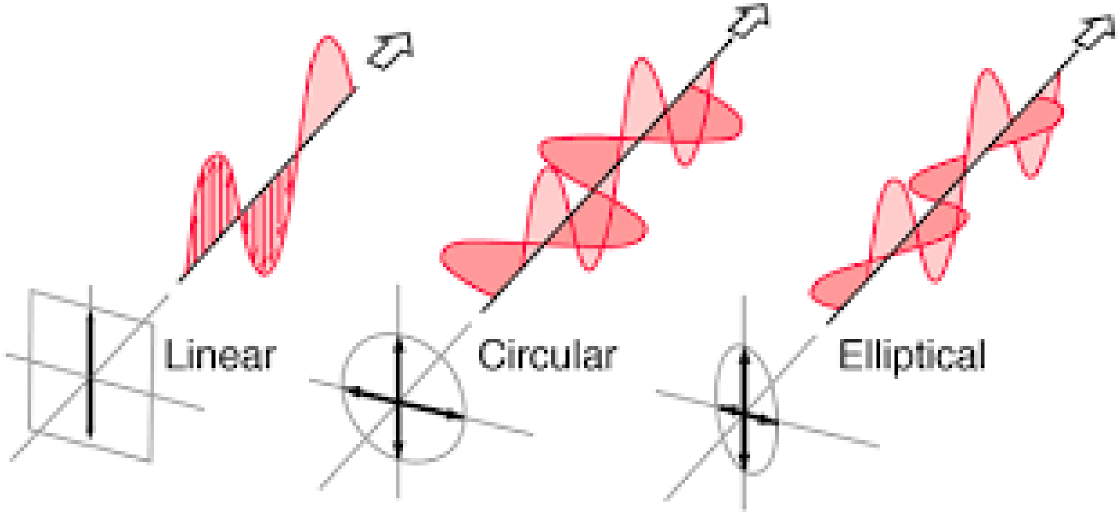
Polarization

Recall Key Concepts:

- EM Waves have two perpendicular fields
- EM Wave polarization is determined by the electric field

New Key Concepts:

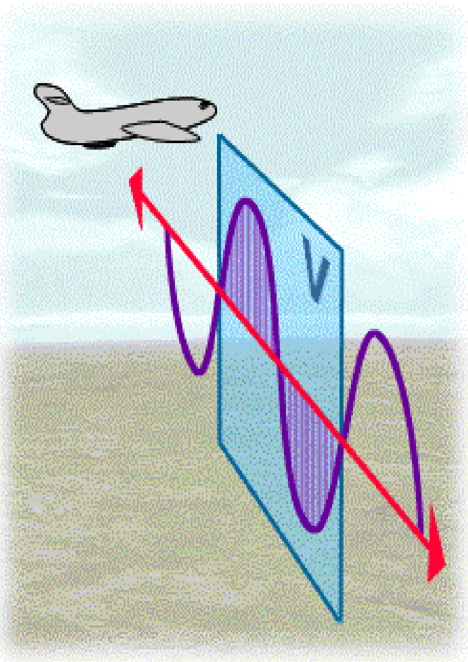
- Antennas are polarized
- Polarization is determined off of the emitted electric field
- Polarization can be
  - Linear
  - Circular
  - Elliptical



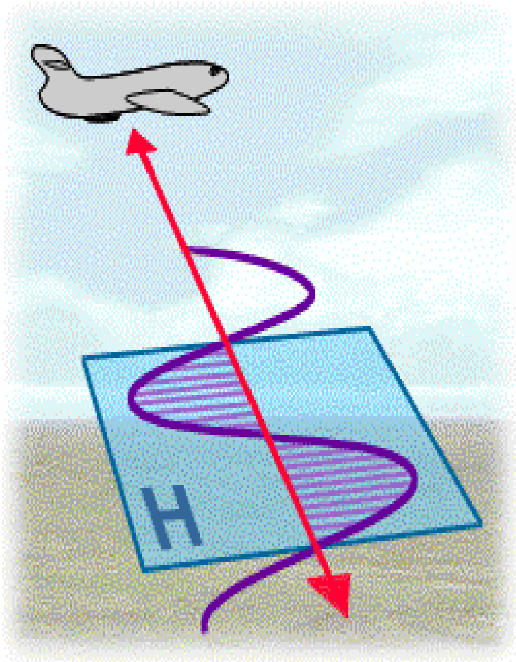
Linear Polarization

Occurs when the electric field and the magnetic field are in fixed perpendicular planes to the path of propagation.

Vertical



Horizontal

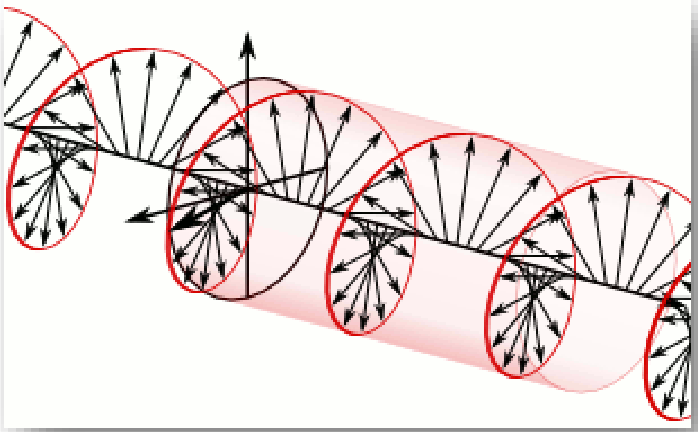


Example

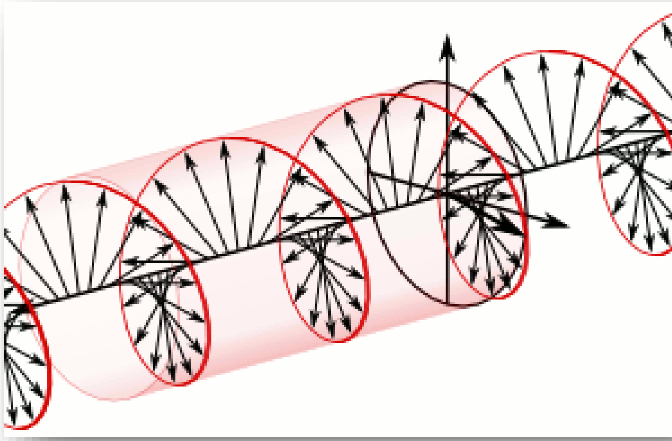
Circular Polarization

Occurs when the electric field and magnetic field rotate around the path of propagation.

Left Hand



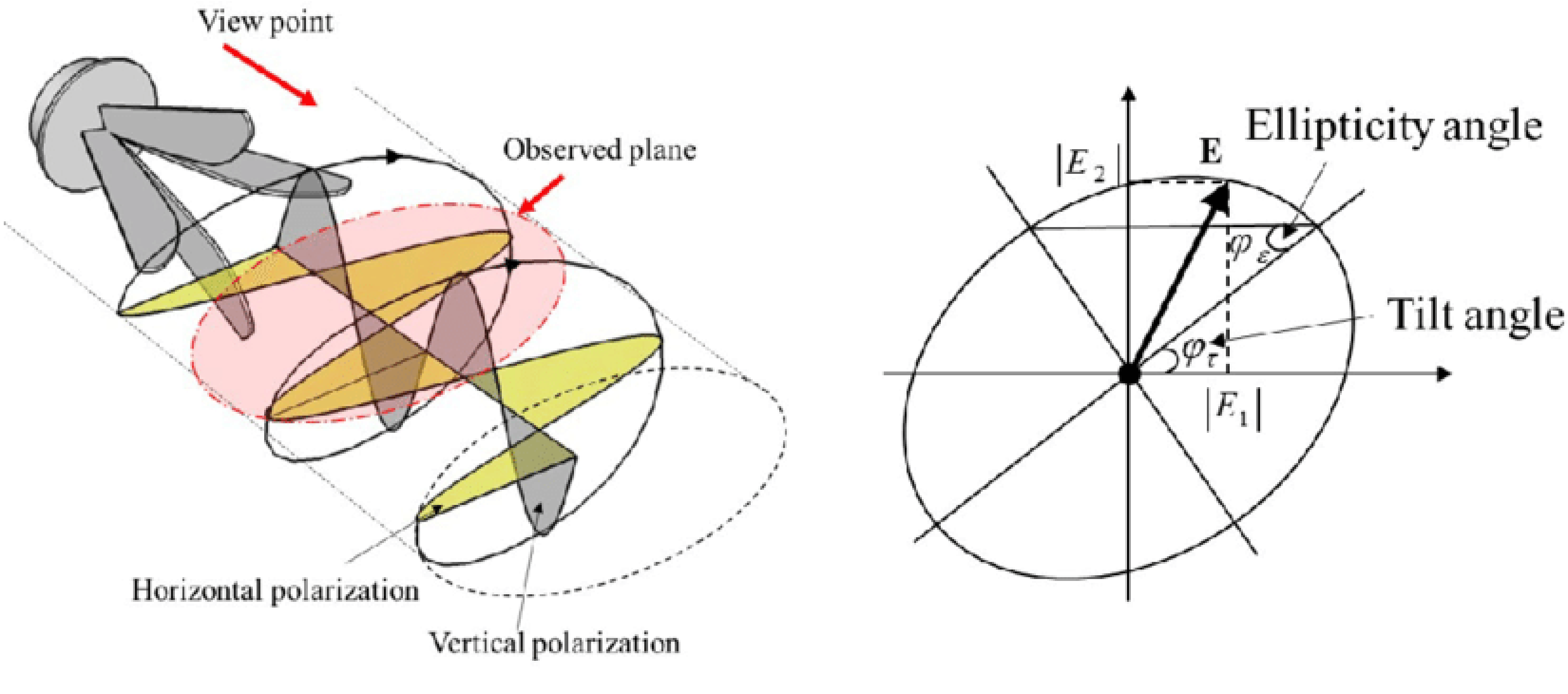
Right Hand



Example

Elliptical Polarization

Occurs when the tip of the electric field vector rotates unevenly around the path of propagation.



(a) (b)

Polarization Applications

Why It Matters:

-Antennas must have the same configuration to have maximum reciprocity between the two

-Vertical antennas will have almost no reciprocity with Horizontal antennas

-Linear antennas will have some reciprocity with Circular antennas

Example:

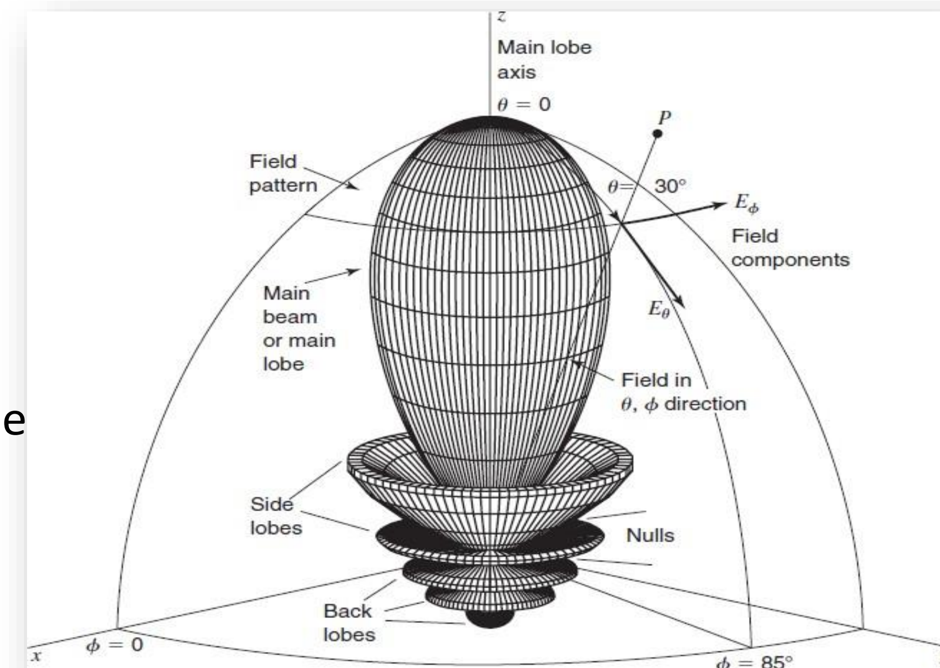
Attempting to use a vertically polarized antenna to communicate with a horizontally polarized antenna is called cross-polarization. The end result is a **30dB** loss (catastrophic!) in signal strength at the receiver.

Remember: Unless designed to do so; if the Rx signal is weaker than the noise floor, typically you will not receive the intended signal.

Of Note! A circular polarized waveform received by a linear polarized antenna will result in a 3dB loss.

Radiation Pattern

- A 3D polar representation of the EM waves emitted from an antenna’s far field (radiation field).
- Determines the coverage and directionality (gain) of the antenna



Polar coordinate system:

ϕ : **Azimuth**: Angle in horizontal plane

(Top down view).

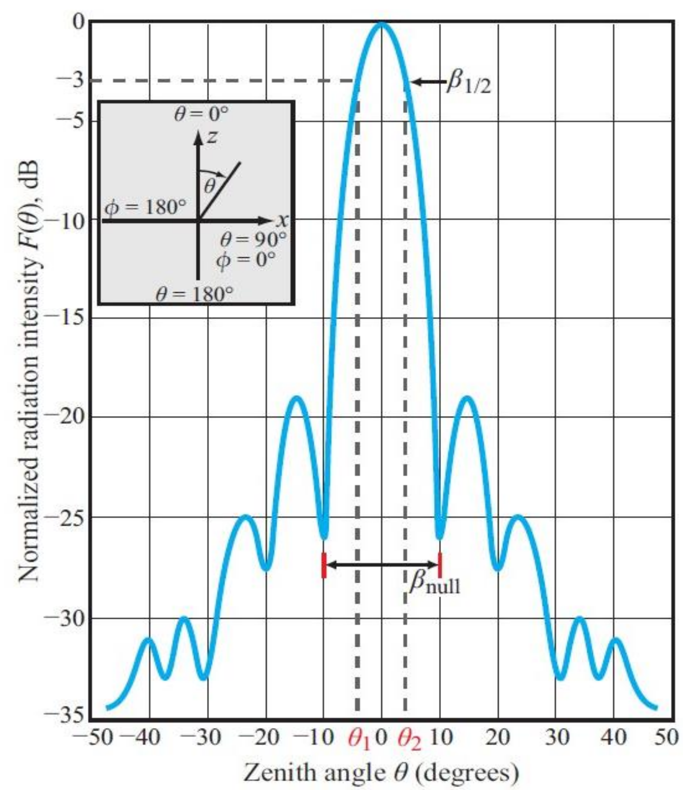
q : **Elevation**: Angle above horizontal

Radiation Characteristics

<u>Main Lobe</u>: The angle and direction at which the maximum radiation occurs**.**

<u>Nulls</u>: Direction and depths of no radiation, 0dB**.**

<u>First Null Beam Width: (HPBW)</u> is the angle of radiated energy between the first nulls in the radiation pattern.

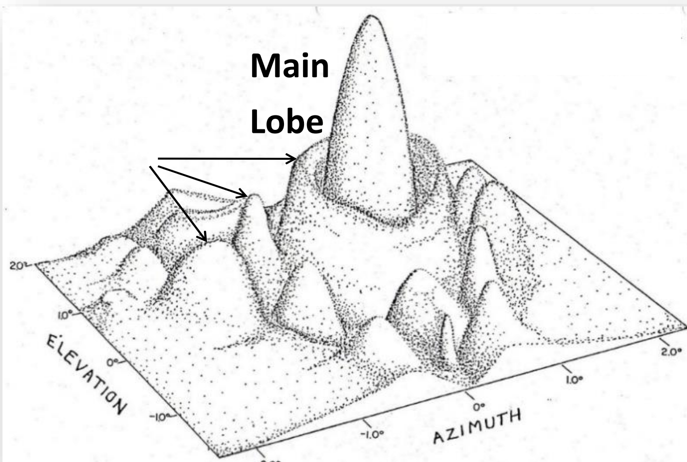


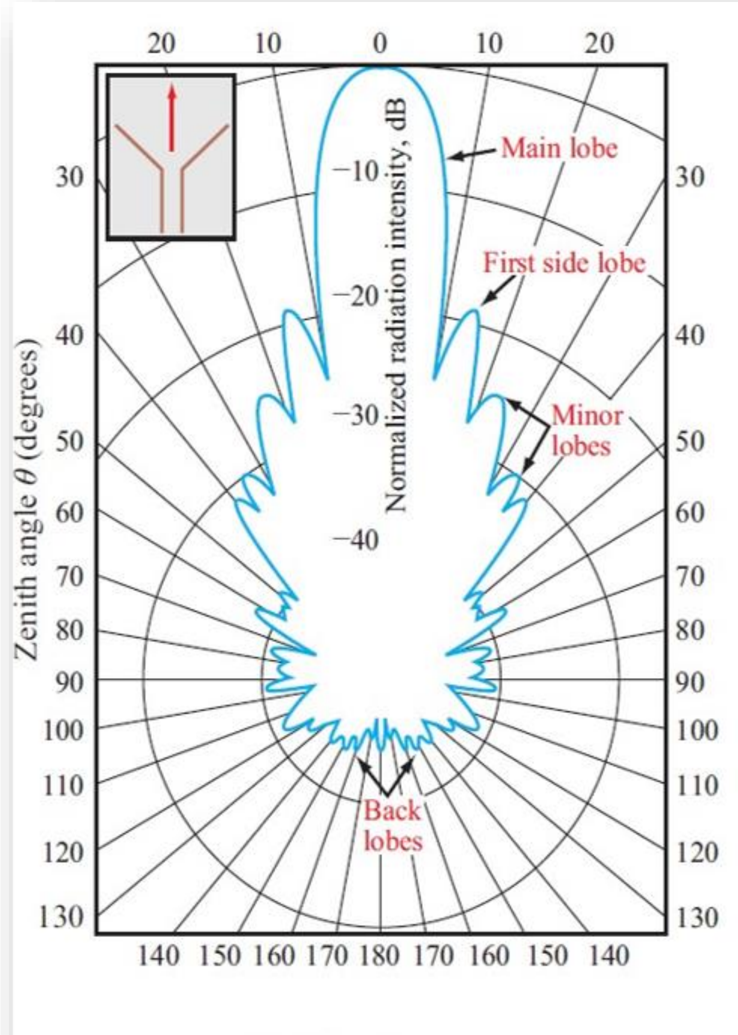
(b) Rectangular plot

Radiation Characteristics

<u>Side Lobes:</u> are the lobes of the radiation pattern outside of the main lobe that are separated by nulls.

<u>Front/Back Ratio: (F/B):</u> the ratio of the maximum signal radiating from the main/front beam to the maximum signal radiating from the back of the antenna.





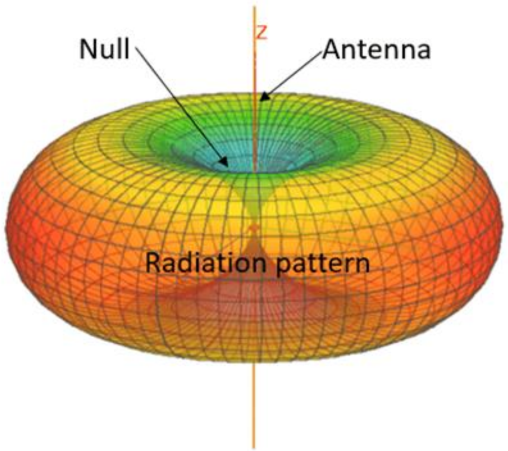
(a) Polar diagram

Directionality (Gain)

UNCLASSIFIED

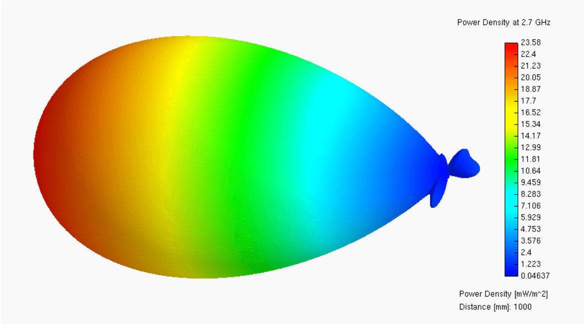
Directionality (Gain):

- The bias of antenna to radiate its energy in a specific direction
- Measured both horizontally and vertically
- Considered as a relative (not real) gain for a radio
- Is measured in Dbi (Decibels Isotropic)



BLUF:

- Significant factor in antenna design and application
- Can greatly increase the gain of a radio
- Can be used In direction finding
- Can be used to defeat interference



Antenna Types

Antenna Shapes:

Wire antennas: Dipole, helix, loop, & Yagi
Aperture antennas: Horn & parabolic dish
Printed antennas: Patch, printed dipole, spiral & slot

Antenna Gain Levels:

High Gain (> 20 dBi): Parabolic dish
Medium Gain (10 to 20 dBi): Horn, helix & Yagi
Low Gain (< 10 dBi): Dipole, loop, patch, slot & whip.

Gain Averages

Type of Antenna	Gain	HPBW
Isotropic	0dBi	360°x360°
Dipole	2dBi	360°x120°
Helix (10 turn)	14dBi	35°x35°
Small Parabolic Dish	16dBi	30°x30°
Large Parabolic Dish	45dBi	1°x1°

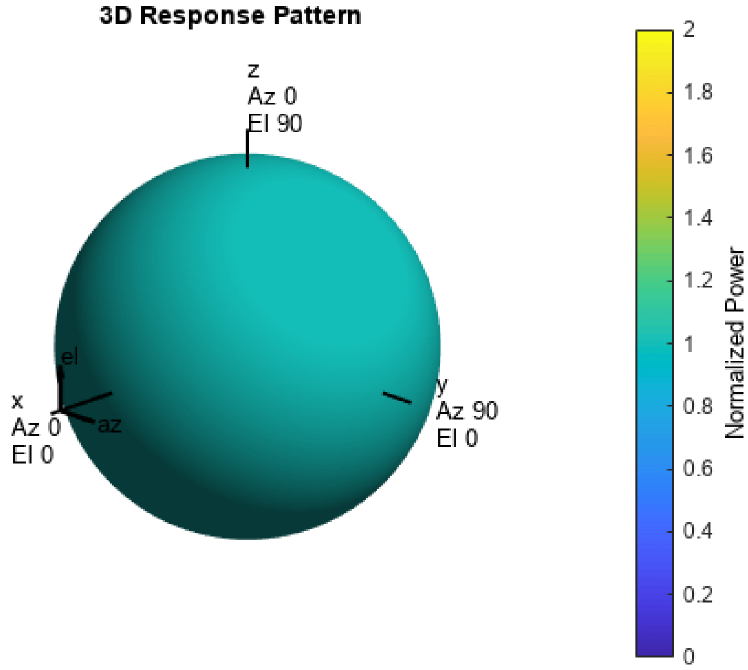
Beam Width Averages

Antenna Type	Horizontal Beam-width	Vertical Beam-width
Omnidirectional	360°	7° to 80°
Patch/Panel	30° to 180°	6° to 90°
Yagi	30° to 78°	14° to 64°
Sector	60° to 180°	7° to 17°
Parabolic Dish	<4° to ~25°	<4° to ~21°

Antenna Types

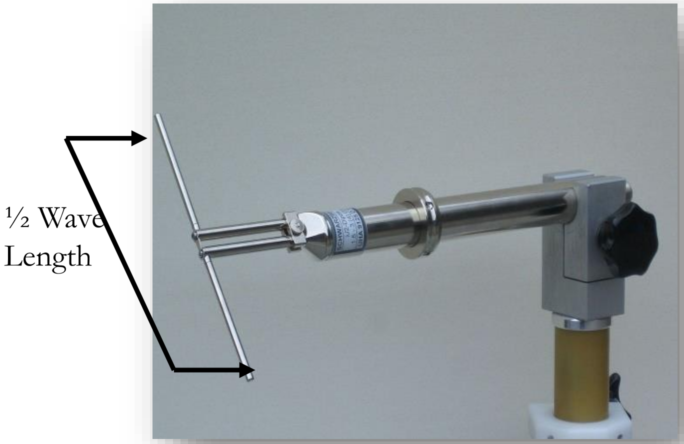
Isotropic: An antenna that emits the same power in all directions

- Theoretical (Does not exist in real life)
- Truly Omni Directional (no nulls)
- Used as benchmark comparison to determine the gain of a real antenna

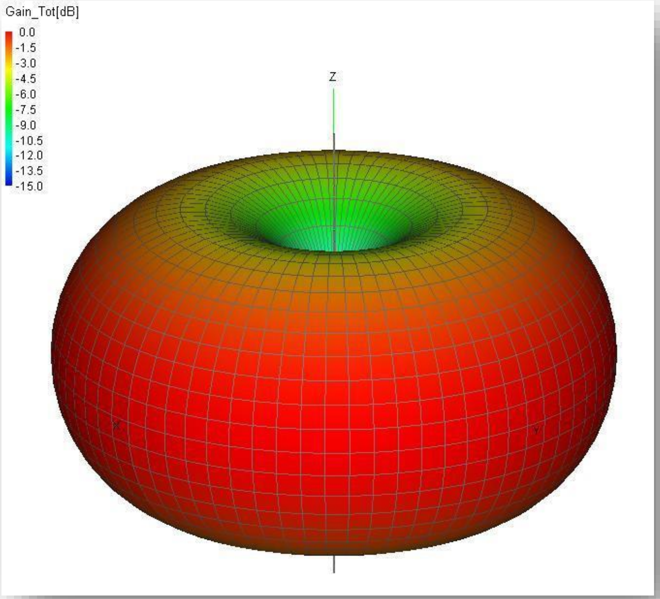


Antenna Types

Half-wave Dipole: (Omni-Directional)



Radiation Pattern



Characteristics:

Polarization: Linear

Directionality: "Omni"

Bandwidth/Gain: Small/~2dBi

Pros: Simple

Works in all directions

Cons: Easily intercepted

Prone to interference

Limited range

Applications:

Wireless Local Area Networks.

VHF TV "Rabbit ears".

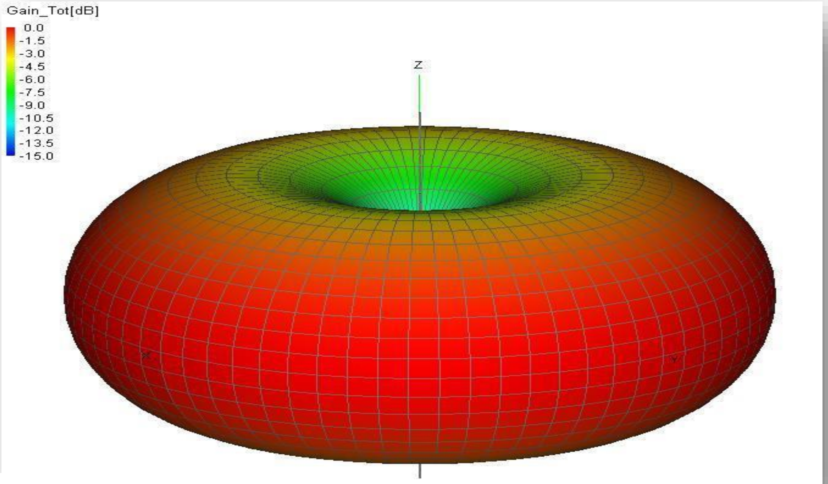
FM radio (folded dipole).

Radio mast transmitters.

Antenna Types

Monopole (Whip): (Omni-Directional)

Radiation Pattern



Characteristics:

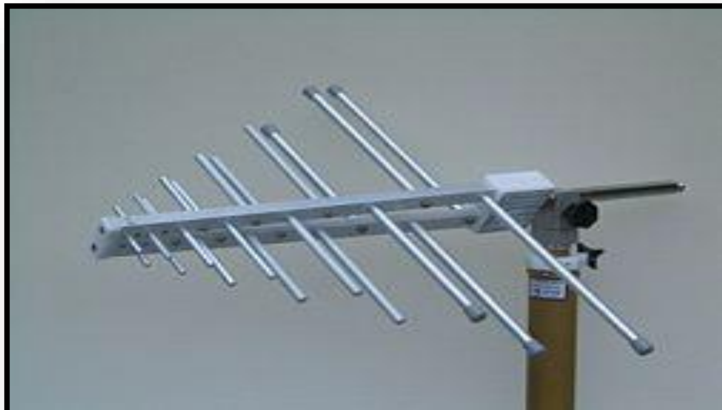
Polarization: Linear
Directionality: "Omni"
Bandwidth/Gain: Small/~2dBi
Pros: Simplest
Works in all directions
Cons: Easily intercepted
Prone to interference
Limited range

Applications:

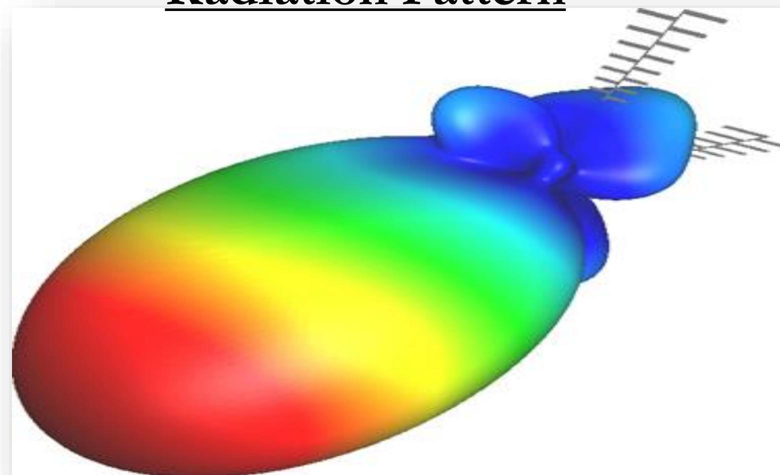
Military Radios (Most Common)
Push to talk radios

Antenna Types

Log Periodic: (Directional)



Radiation Pattern



Characteristics:

Polarization: Linear

Directionality: Moderate to High gain

Bandwidth/Gain: large/5-10dBi

Pros: Large bandwidth of frequencies
Good gain over bandwidth

Cons: Complicated to construct
Does not work in all directions
Large apparatus generally required

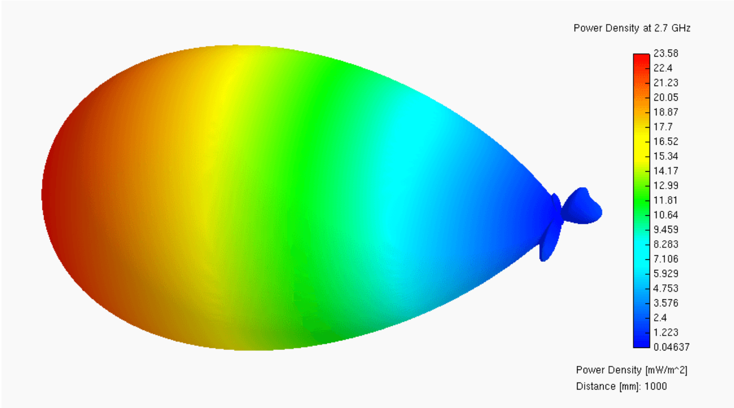
Applications:

Spectrum Analyzer's
Interference Hunting
TV reception

Antenna Types

YAGI: (Directional)

Radiation Pattern



Characteristics:

Polarization: Linear

Directionality: Moderate to High gain

Bandwidth/Gain: Small-Medium/5-10dBi

Pros: Great gain on specific frequencies
Easier to construct than LPA

Cons: Smaller bandwidth than LPA

Applications:

Direction Finding

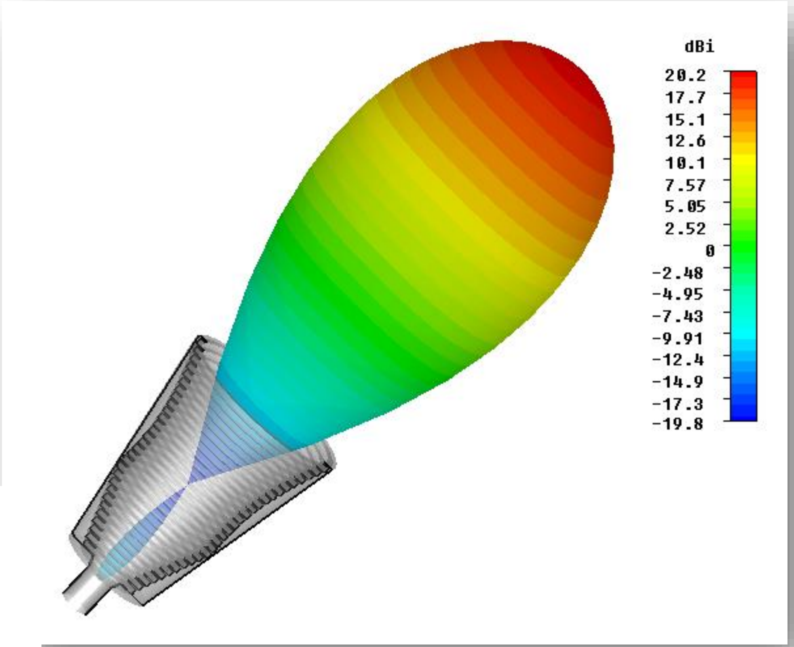
Extended LOS coms

SCADA Systems

Antenna Types

Conical Spiral: (Directional)

Radiation Pattern



Characteristics:

Polarization: Circular

Directionality: High gain

Bandwidth/Gain: Small/8+dBi

Pros: High fidelity signal
Hard to intercept or interfere with

Cons: Small coverage radius
Must know location of receiver
Small bandwidth

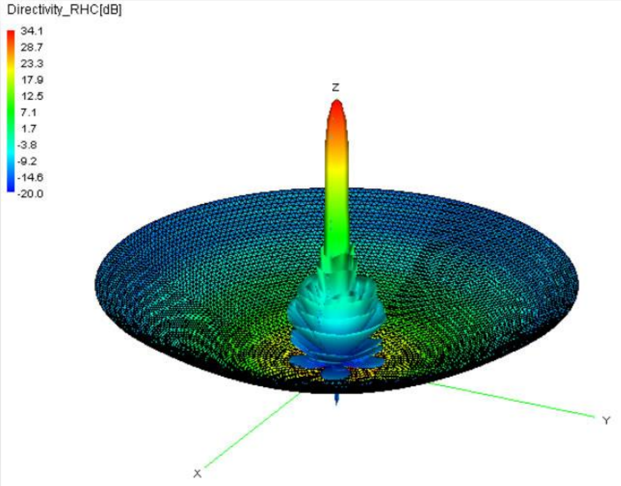
Applications:

Point to Point Communications and Microwave links

Antenna Types

Parabolic Dish: (Extremely Directional)

Radiation Pattern



Characteristics:

Polarization: Depends on feed.

Directionality: Extremely high gain

Bandwidth/Gain: Small/10+ dBi

Pros: Great reception of specific signals
Not prone to general interference
Can receive and transmit at long distances

Cons: Generally large apparatus
Small coverage radius (must be in direction of main lobe)

Applications:

Satellite Communications

Point-to-Point Backhaul.

- Cellular telephony, Wi-Fi

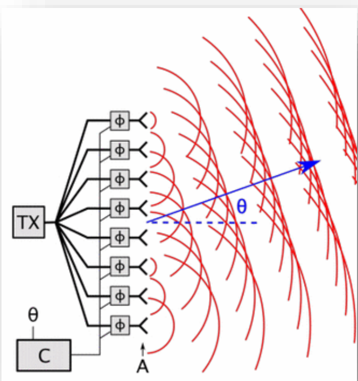
Antenna Types

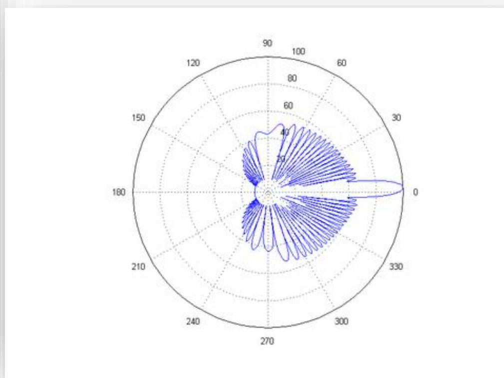
Phased Array (Fan Beam):

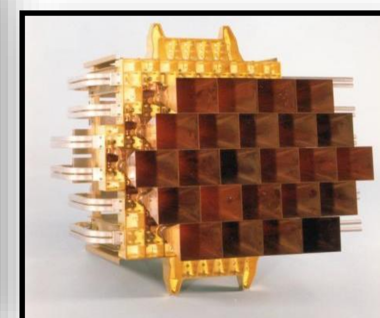
Is an electronically controlled array of antennas that creates a beam of radio waves and can be steered to point in different directions, without moving the antennas.

How it works:

The power from the transmitter is fed to the antennas through devices called phase shifters, which alter the phase electronically. (Used heavily in military radar technology).







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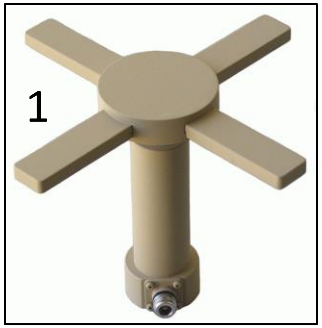
Check on Learning



Questions?